

TAMEDIA OFFICE BUILDING

SHIGERU BAN ARCHITECTS.
2013
ZURICH, SWITZERLAND



The Tamedia Office Building in Zurich is a seven-story headquarters for the Swiss media company Tamedia, completed in 2013. It is notable as the country's tallest timber-construction building at the time and the first Swiss project by architect Shigeru Ban . Celebrated for its revolutionary all-wood structure, "Shigeru Ban's design for the seven-story Tamedia office building in Zürich raises the bar with its unique timber structure" . The following report examines the building's architectural design, structural engineering, material choices, sustainability considerations, and other distinctive features.

ARCHITECTURAL DESIGN

Tamedia's headquarters presents a modernist glass volume that highlights its structure as a central aesthetic element. The design draws inspiration from traditional Japanese carpentry: the building's wooden frame is entirely interlocking and devoid of joint hardware and glue, creating an elegant lattice visible behind a low-e coated glass facade . Although contemporary in construction, the building sensitively responds to its historic context. The glazed cube is topped by a two-story mansard roof, a vernacular form prevalent in the surrounding Verdareal district . This roof profile helps the new structure blend with neighboring buildings while concealing rooftop mechanicals within a shape common to the area's architecture. At street level, a generous double-height lobby echoes the scale of adjacent buildings, making the entrance inviting and in harmony with the streetscape .

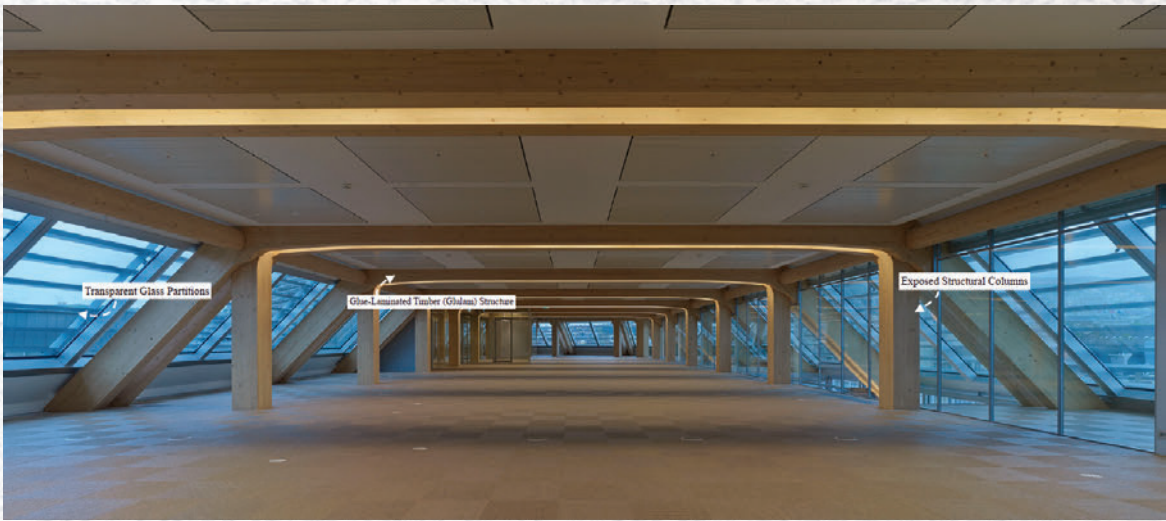
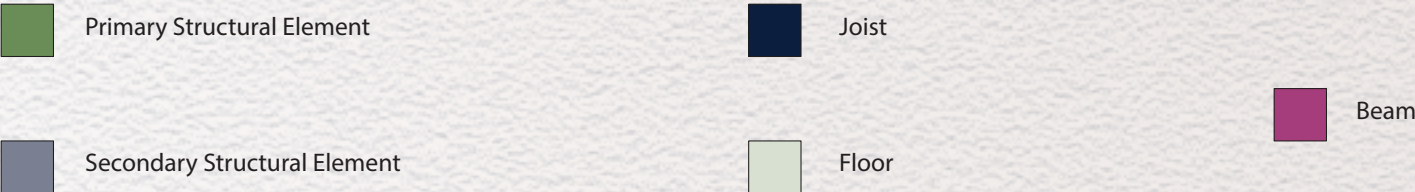
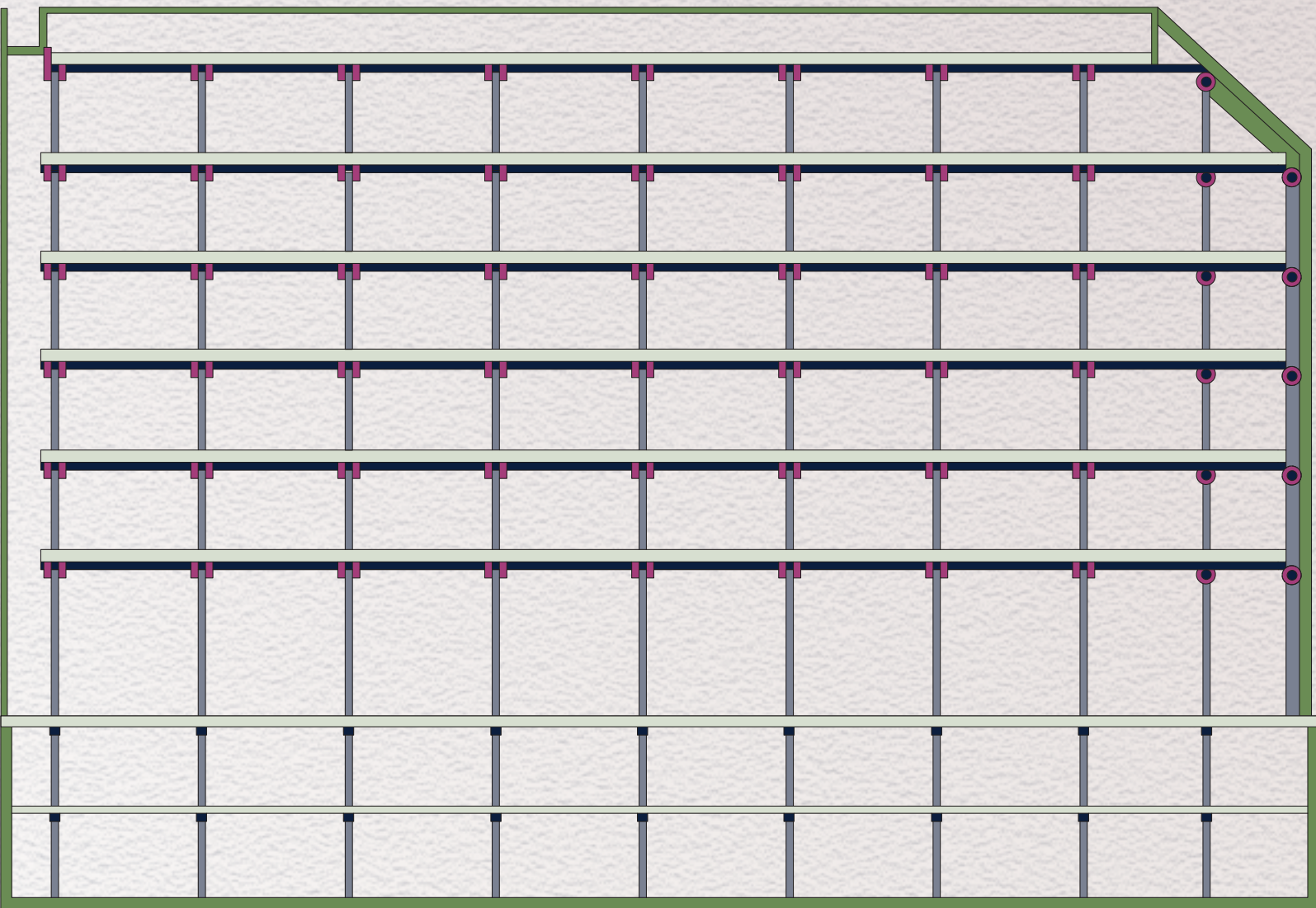
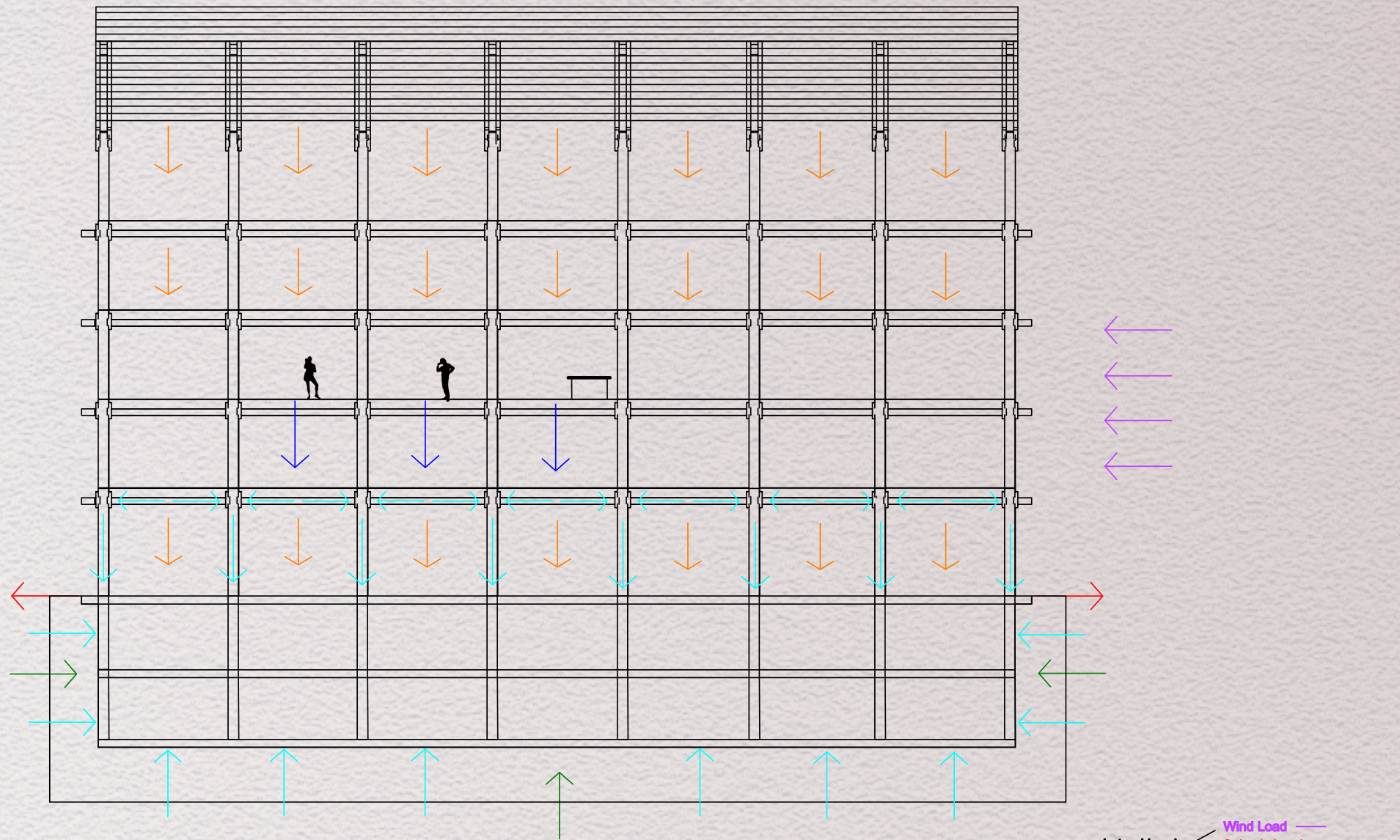
ENGINEERING AND SUSTAINABILITY

The primary structure consists of a grid of massive glued-laminated spruce columns and beams configured as seven portal frames spanning the building's width. Each frame rises roughly 21 meters and weighs on the order of 20 to 24 tonnes. All the connections between the wood members are formed without metal fasteners. Instead of steel plates or bolts, the beams and columns are engineered to interlock and are secured by hardwood dowels, eliminating the need for any mechanical connectors or adhesives. High-precision CNC milling was used to fabricate these connections, ensuring the beech wood pins fit perfectly and the entire frame remains rigid and stable under load. The exterior of the building is wrapped in a fully glazed curtain wall that puts the timber interior on display. The glass has a low-emissivity coating and high thermal performance, meeting Switzerland's stringent energy codes while maintaining transparency . In fact, the facade is a double-skin system: an outer glass layer and an inner glass layer create an insulated air cavity in between (visible as the "intermediate space" along the canal side). This double-glass envelope helps regulate temperature and sound, and its floor-to-ceiling clarity maximizes daylight and views for the occupants. The sleek glass skin paired with the warm wood structure gives the building its distinctive blend of modernity and natural materiality. The choice of a timber structural frame greatly reduced the building's embodied energy and carbon footprint compared to a conventional concrete or steel structure. By using 2,000 m³ of sustainably harvested spruce, the Tamedia project significantly cut the "grey energy" (embodied energy) of the building. This approach aligned with the city of Zurich's forward-looking environmental policies – in 2008 the local population adopted the 2000-Watt Society goal, aiming to drastically reduce per-capita energy use.

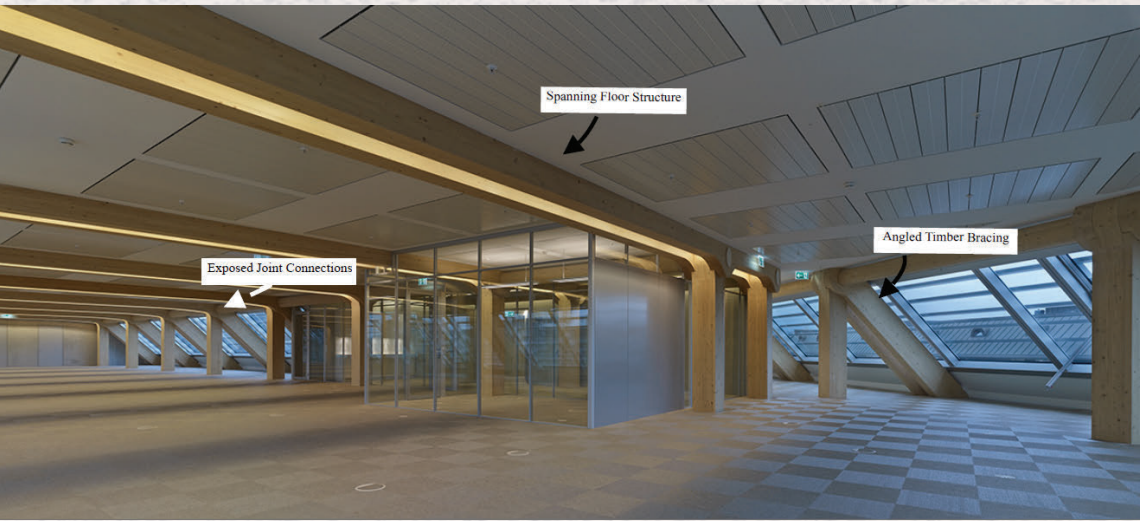
The building also excels in energy efficiency and low-impact operation. Its HVAC system utilizes geothermal groundwater wells for heating and cooling, eliminating reliance on oil or gas boilers. A heat pump uses the constant temperature of groundwater to provide climate control, which "negates the use of fossil fuels entirely" for the building's heating/cooling needs . The design of the facade further reduces energy demand: the double-skin glass facade creates an insulating buffer zone that pre-temperes the air, cutting down on heat loss in winter and heat gain in summer . This cavity, along with high-performance glazing, allows the building to meet Switzerland's strict energy codes and maintain comfortable indoor climates with minimal active conditioning. Additionally, many of the facade panels are operable (retractable), so that in mild weather the office floors can be naturally ventilated with fresh air . By opening these large glass sections, meeting rooms and workspaces can effectively become open-air terraces, reducing the need for air conditioning and enhancing user comfort. All these features mean the Tamedia headquarters consumes very little energy for its size – its operation produces virtually no CO₂ emissions, with no fossil fuel use and even no dependence on non-renewable electricity sources (the heat pumps run on grid electricity, which in Switzerland is largely free of coal/gas and even avoids nuclear power) . In sum, the building achieves an almost carbon-neutral performance in day-to-day use, far surpassing typical standards and serving as a model for environmentally sustainable office design .

REFERENCES:

- Blumer, Hermann. Timber Engineering and Construction Techniques in Contemporary Architecture. Springer, 2015.
- Lehmann, Stefan. Wood as a Sustainable Construction Material: Carbon Sequestration in Architecture. Routledge, 2018.
- Schmid, Martin. "Zurich's First Large-Scale Timber Office: Structural Innovations in the Tamedia Headquarters." Journal of Timber Engineering, vol. 24, no. 3, 2014, pp. 56-75. Tamedia AG. Tamedia Headquarters Zurich: Architectural and Structural Report. Tamedia, 2013.
- Ban, Shigeru. "Tamedia Office Building: A New Era of Sustainable Timber Architecture." Shigeru Ban Architects, 2013, www.shigerubanarchitects.com.
- "Tamedia Media Headquarters in Zurich: Swiss Timber Innovation." Swiss Architecture Review, 14 Mar. 2014, www.swissarchitecture.ch/tamedia.
- "Zurich's Tallest Timber Building: The Tamedia Office Project." ArchDaily, 21 Nov. 2013, www.archdaily.com.
- Jodidio, Philip. "The Fusion of Tradition and Innovation in Tamedia's Office Building." The Architectural Review, vol. 239, no. 5, 2014, pp. 42-58.
- "Building with Timber: Tamedia Headquarters, Zurich." Swiss Timber Journal, issue 32, 2014, pp. 18-24.
- Schuler, Robert. "Revolutionizing Office Spaces: Inside the Tamedia Building." Journal of Green Building, vol. 9, no. 2, 2015, pp. 35-48.



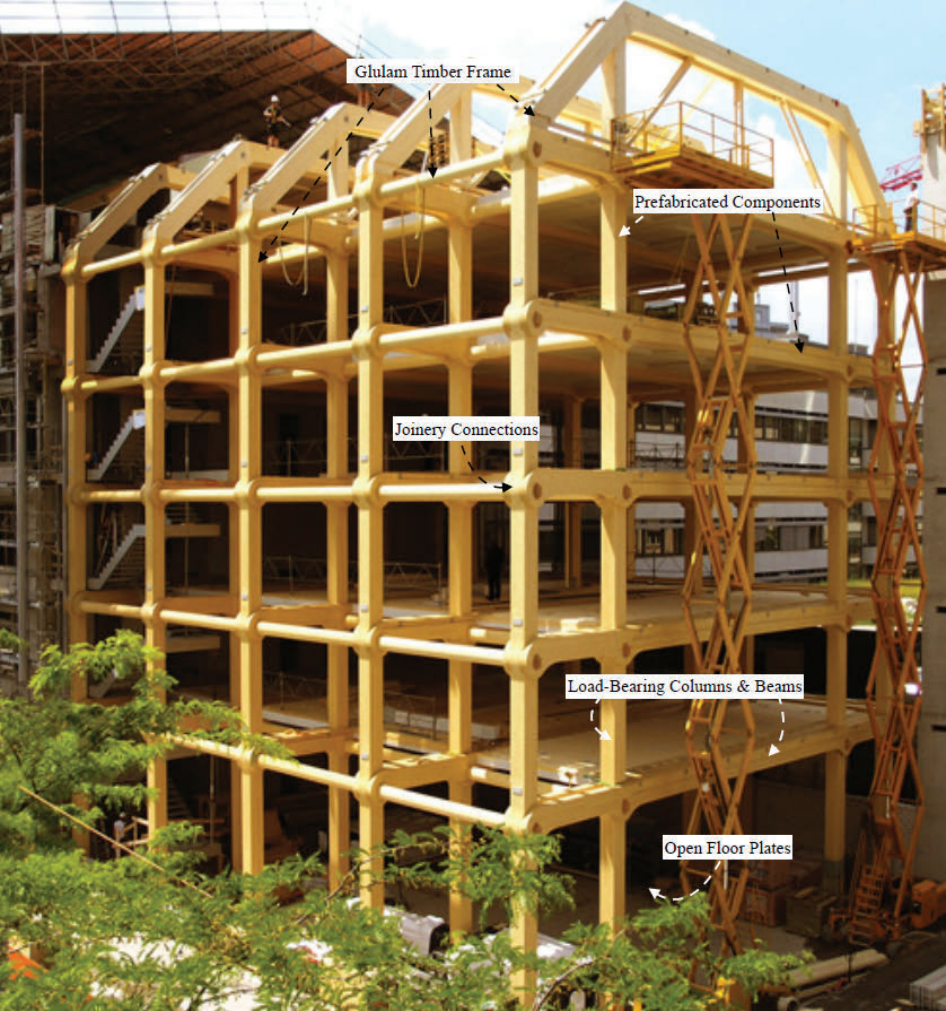
Interior view showcasing the prefabricated glulam timber system, with exposed structural columns and beams providing open-plan flexibility. The use of glass partitions maximizes natural light and promotes clear spatial organization, reflecting Shigeru Ban's commitment to sustainable and transparent design.



This interior view highlights the building's spanning floor structure, which supports the loads between timber columns. The angled timber bracing provides lateral stability along the glazed facade, while the exposed joint connections demonstrate the precision and craftsmanship of the timber construction, eliminating the need for steel reinforcements.



This exterior perspective highlights the building's modular structural grid system, supported by a glulam timber frame. The transparent glass curtain wall reveals the exposed timber framing, emphasizing Shigeru Ban's sustainable and honest material design approach.

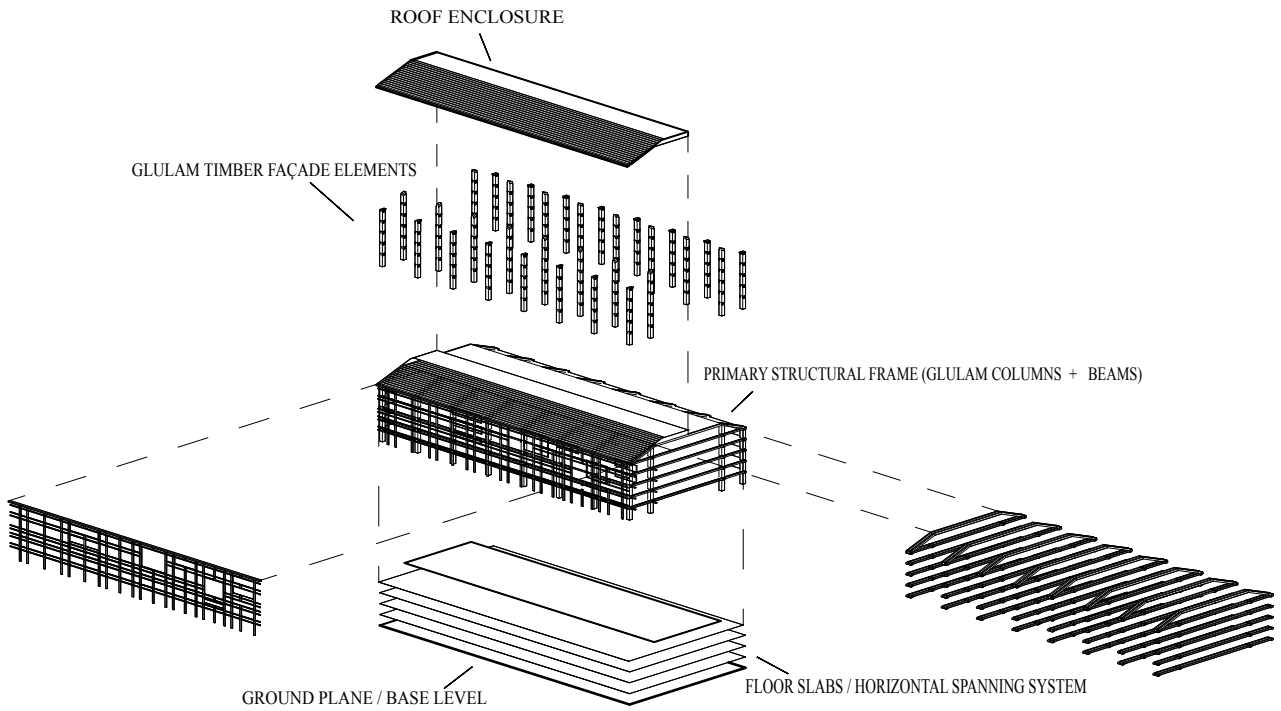


This construction phase view of the Tamedia Building reveals its precision-engineered glulam timber frame, assembled from prefabricated components. The load-bearing columns and beams create open floor plates, allowing flexible interior layouts. Exposed joinery connections demonstrate advanced craftsmanship. Together, these elements reflect Shigeru Ban's sustainable design approach, emphasizing transparency and efficiency.



This image highlights the precision-crafted timber joinery used in the Tamedia Office Building. The interlocking wooden elements efficiently transfer structural loads between columns and beams.

EXPLODED AXONOMETRIC DIAGRAM
TAMEDIA OFFICE BUILDING - SHIGERU BAN ARCHITECTS



LEGEND	
	Primary Structural Frame (Glulam Columns + Beams)
	Floor Slabs / Horizontal Spanning System
	Exploded Guideline